METRIC
MIL-S-24700(SH)
12 February 1988
SUPERSEDING
MIL-S-23920(SHIPS)
14 November 1963
(See 6.5)

MILITARY SPECIFICATION

STUFFING TUBES FOR REFERENCE ELECTRODES, IMPRESSED CURRENT CATHODIC PROTECTION (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

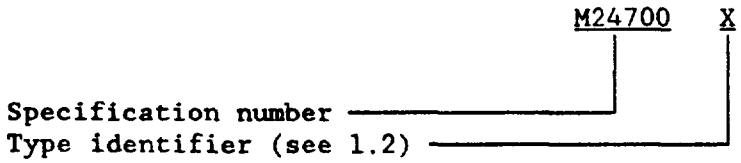
1. SCOPE

1.1 Scope This specification covers the requirements for stuffing tubes to be used for attachment of reference electrodes to ship hulls or similar structures for corrosion protection by impressed current cathodic protection. It includes required features of fabrication of the stuffing tubes.

1 2 Classification The stuffing tubes shall be of the following types as specified (see 6.2 1):

Type I - Reference electrode stuffing tube for surface ships
Type II - Reference electrode stuffing tube for submarines

1.3 Part numbers Part numbers under this specification shall be formulated as follows (see 3 4.1 and 6.2):



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-N-281	- Nickel-Copper Alloy Bar Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-S-365	- Silver Plating, Electro-deposited: General Requirements for
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-C-915	- Cable and Cord, Electrical, for Shipboard Use, General Specification for.
MIL-C-915/6	- Cable, Electrical, 600 Volts, Types SHOF, DHOF, THOF, and FHOF.
MIL-C-915/8	- Cable, Electrical, 600 Volts, Types DSS, TSS, FSS and 7SS
MIL-S-16974	- Steel Bars, Billets, Blooms and Slabs Carbon and Alloy (for Reforging or Other Operations Before Heat Treatment).
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated
MIL-P-19468	- Plastic Rods, Polytetrafluoroethylene, Molded and Extruded
MIL-S-22698	- Steel Plates and Shapes, Weldable Ordinary Strength and Higher Strength: Hull Structural
MIL-S-23284	- Steel Forgings, Carbon and Alloy, for Shafts, Sleeves, Couplings, and Stocks (Rudders and Diving Planes).
MIL-E-23919	- Electrode, Reference, Circular, Corrosion Preventive (Flush Type): Silver-Silver Chloride
MIL-S-24093	- Steel Forgings, Carbon and Alloy Heat Treated.
MIL-C-24231	- Connectors, Plugs, Receptacles, Adapters, Hull Inserts, and Hull Insert Plugs, Pressure Proof, General Specification for.
MIL-C-24231/25	- Connector, Hull Inserts

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services

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MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging Requirements.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 29 - Standard Specification for Steel Bars, Carbon and Alloy, Hot-Wrought and Cold-Finished, General Requirements for (DoD adopted)

B 139 - Standard Specification for Phosphor Bronze Rod, Bar, and Shapes. (Metric) (DoD adopted)

D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials. The stuffing tube shall consist of a metal body to receive the projecting hub and pin of a reference electrode conforming to MIL-E-23919 and to hold the end of the pin in contact with an electrical cable that will transmit current to other components of the impressed current cathodic protection system

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3.2.1 Body.

3.2.1.1 Type I. The body shall be steel conforming to grade 1018 of ASTM A 29. An alternative material shall be steel conforming to number 8620 of MIL-S-16974. On hulls where this material will not be compatible due to welding or similar effects, (such as, aluminum), an alternative material shall be specified (see 6.2.1).

3.2.1.2 Type II. The body shall be steel conforming to grade AH-36 or grade AH-36T of MIL-S-22698. An alternative material shall be steel conforming to number 8620 of MIL-S-16974 that has been heat-treated to obtain the physical properties of steel conforming to class F of MIL-S-24093 or class 2 of MIL-S-23284

3.2.2 Bushing The bushing used to insulate and center the reference electrode pin inside the stuffing tube body shall be polytetrafluoroethylene (PTFE) conforming to group B of MIL-P-19468.

3.2.3 Connector assembly Each connector assembly shall consist of a connector tube and a compression ring in a single piece. Stuffing tubes shall have connector assemblies fabricated of silver-plated phosphor-bronze The silver plating shall conform to type I, grade A of QQ-S-365 and shall have an average thickness of 13 microns, minus 0 microns plus 13 microns. The phosphor-bronze shall be an alloy conforming to UNS no. C54400 of ASTM B 139.

3.2.4 Molded packing. The molded packing shall be a watertight seal. The molded packing shall have the formulation specified in table I.

TABLE I. Formulation for molded packing for stuffing tubes.

Component	Part by weight
Geon polyblend 503 (or equivalent)	100
MPC carbon black (or equivalent)	5
Stearic acid	1
Dibutyl phthalate	20
Altax or MBTS (or equivalent)	1.5
Litharge	10
Crystex (insoluble sulfur) (or equivalent)	1 5

3.2.5 Gland nut assembly The components of the gland nut assembly (gland nut, gland ring, and spring-lock washer) shall be monel metal conforming to QQ-N-281

3.2 6 Connecting cable. The connecting cable shall be designation DSS-3 conforming to MIL-C-915 and MIL-C-915/6 or MIL-C-915/8. The type and length of cable shall be as specified (see 6 2 1).

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the gland nut shall be 1 11 UNF 2A in accordance with FED STD H28.

3.3.6 Eccentricity of body cavity. The cavity of the steel body that accepts the hub of the reference electrode shall be positioned axially and concentrically in relation to the outer diameter. Eccentricity shall be not more than 127 microns (0.005 inch).

3 3.7 Hull insert. The hull insert shall be constructed in accordance with MIL-C-24231, except that it shall meet the dimensional requirements of the stuffing tube specified herein.

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3.4 Marking of stuffing tubes

3.4.1 Manufacturer's symbol A manufacturer's identifying symbol shall be stamped or engraved on the outer surface of the body. A capital letter corresponding to the revision number of the Military specification to which the stuffing box conforms shall also be used in addition to the manufacturer's symbol. Type II stuffing tubes shall be marked to indicate submarine service The hull insert for type II stuffing tubes shall have identical markings.

3.5 Workmanship. The surface of all components of the stuffing tube shall be smooth and free of burrs, inclusions, dents, scratches, cuts, and other defects of similar nature.

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.1.1 Responsibility for compliance All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system When specified in the contract or order an inspection system program plan shall be prepared (see 6 2 2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4 3 First article inspection Stuffing tubes shall meet the first article inspection requirements of MIL-C-24231, except that the destructive bonding, high potential, and hardness tests are not required Hydrostatic pressure testing shall be accomplished without a MIL-E-23919 reference electrode installed in place in the stuffing tube The stuffing tube shall be installed in a hull plate model and pressure applied to the reference electrode side.

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4.3.1 Sampling for first article inspection. One sample shall be subjected to the first article tests specified in 4.3.

4 3.2 First article inspection report. When specified in the contract or order, a first article inspection report shall be prepared (see 6.2.2).

4.4 Quality conformance inspection. Stuffing tubes and hull inserts shall meet the quality conformance testing requirements of MIL-C-24231, except the packing hardness test is not required. Hydrostatic pressure tests shall be accomplished with a MIL-E-23919 reference electrode in place in the stuffing tube The stuffing tube shall be installed in a hull plate model and pressure applied to the reference electrode side.

4.4.1 Quality conformance inspection report. When specified in the contract or order, a quality conformance inspection report shall be prepared (see 6.2.2).

4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5 1 General.

5.1.1 Navy fire-retardant requirements.

5.1.1.1 Lumber and plywood. When specified (see 6.2.1), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B	-	Type II - weather resistant. Category I - general use.
Level C	-	Type I - non-weather resistant. Category I - general use.

5.1.1.2 Fiberboard. When specified (see 6.2.1), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes shall meet the flame spread index and specific optic density requirements of PPP-F-320 and amendments thereto

5.2 Preservation Preservation shall be level A, C, or commercial as specified (see 6.2 1).

5.2.1 Level A. Unless otherwise specified (see 6.2.1), tubes shall be unit packed one to a fiberboard box. Tubes shall be immobilized within the boxes with fiberboard pads, die-outs or separators to obtain a snug fitting nonshifting

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load. The fiberboard box and inserts shall conform to PPP-B-636, class-weather resistant. The box closure and reinforcing shall be in accordance with the requirements for method V of the appendix to the box specification.

5.2.2 Level C. Preservation shall be as specified under level A except that the fiberboard boxes shall conform to the class-domestic and closed in accordance with method I using pressure sensitive tape. When specified (see 6.2.1), fiberboard boxes shall conform to class-domestic/fire retardant.

5.2.3 Commercial. Commercial preservation shall be in accordance with the requirements of ASTM D 3951.

5 3 Packing Packing shall be level A, B, C, or commercial as specified (see 6 2.1)

5 3 1 General requirements for levels A, B, and C. Containers selected (see 5.3.2) shall be of minimum weight and cube consistent with the protection required, of uniform size, and contain identical quantities of identical types of stuffing tubes.

5 3 2 Levels A, B, and C containers. Stuffing tubes preserved as specified (see 5 2) shall be packed in exterior shipping containers for the level of packing specified (see 5.3), in accordance with table VII of MIL-STD-2073-1, appendix C, and herein. Unless otherwise specified (see 6.2.1), container selection shall be at the contractor's option. For levels B and C packing, tubes unit packed in class-weather or nonweather resistant, or class-domestic/fire retardant boxes will not require over packing.

5.3.2 1 Caseliners, closures and gross weight

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2.1), level A shipping containers containing tubes preserved level C or commercial shall be provided waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic or fire retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5 3 2 1.3 Weight. Wood, plywood, and cleated type containers exceeding 90.72 kg (200 pounds) gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3 3 Commercial. Tubes preserved as specified (see 5.2) shall be packed for shipment in accordance with ASTM D 3951 and herein

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5.3.3.1 Container modification. Shipping containers exceeding 90.72 kg (200 pounds) gross weight shall be provided with a minimum of two, 75- by 100-mm (3- by 4-inch) nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, storage, and stowage.

5.4 Palletized unit loads. When specified (see 6.2.1), containers shall be palletized in accordance with MIL-STD-2073-1.

5.5 Marking.

5.5.1 Levels A, B, C, and commercial. In addition to any special marking required (see 6.2.1) and 3.4 herein, interior (unit) packs, shipping containers and palletized unit loads shall be marked for shipment, stowage, and storage in accordance with MIL-STD-2073-1, appendix F.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Type I stuffing tubes are intended for use in corrosion prevention in sea water of ship hulls, steel and aluminum equipment and structures, seachests, sonar domes, and similar marine structures when used in conjunction with silver-silver chloride reference electrodes.

6.1.2 Type II. Type II stuffing tubes are intended for use in corrosion prevention in sea water of submarine hulls when used in conjunction with silver-silver chloride reference electrodes.

6 2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following for type I-reference electrode stuffing tube for surface ships:

(a) Title, number, and date of this specification.
(b) Type of stuffing tube and alternative body material if required (see 1.2 and 3.2.1.1).
(c) Type and length of cable attached to the stuffing tube (see 3.2.6).
(d) When fire-retardant requirements are required (see 5.1.1 and 5.2.2).
(e) Level of preservation, packaging, and packing required (see 5 2 and 5.3).
(f) Number of stuffing tubes per package (see 5.2.1).
(g) Container selection if other than contractor's option (see 5.3.2).
(h) When caseliners are not required (see 5.3.2.1.1).
(i) When palletization is required (see 5.4).
(j) Special marking required (see 5.5).

6.2.1.1 Acquisition documents for type II-reference electrode stuffing tube for submarines should conform to MIL-C-24321.

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6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.1 2	Inspection system program plan	DI-R-4803	----
4.3.2	First article inspection report	DI-T-4902	----
4.4 1	Inspection and test reports	DI-T-5329	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6 2.2.1 The data requirements of 6 2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit of each type The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6 4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped

6 5 Supersession data. This specification includes the requirements of MIL-S-23920(SHIPS) dated 14 November 1963

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6.6 Subject term (key word) listing.

Corrosion protection
Explosionproof
Seal
Silver-silver chloride
Sonar domes

Preparing activity:
Navy - SH
(Project 5975-N627)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7
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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-24700(SH)	12 February 1988

3. DOCUMENT TITLE
Stuffing Tubes For Reference Electrodes, Impressed Current Cathodic Protection (Metric)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Mr. Meikle (POC)(Technical)	(703) 602-0214	332-0214
c ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (Code 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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